

MATH3303 – Lecture 11

Classification of Finite Abelian Groups II: p -Groups

Review Notes

Reference: Lee, Abstract Algebra, §5.1–5.2

Overview

- **Lemma 2** (Lee, Thm. 5.2): If G is the internal direct product of $N_1, \dots, N_k$, then $G \cong N_1 \times \dots \times N_k$ (external). Full proof using the commuting lemma.
- **Example:** $\mathbb{Z}_{15} \cong \mathbb{Z}_3 \times \mathbb{Z}_5$ via internal direct product.
- **p -groups:** p -elements, p -groups, and the Cauchy characterisation (Lee, Def. 5.3, Thm. 5.5, Cor. 5.2).
- **Lemma 3** (Lee, Lemma 5.3): p -elements form a subgroup in any abelian group.
- **Lemma 4** (Lee, Lemma 5.4): every element of finite order in any group is a product of p -elements.

1 Internal $\cong$ External Direct Product

Theorem 1.1: Internal direct product is isomorphic to external (Lemma 2 from lecture; cf. Lee, Thm. 5.2)

If G is the internal direct product of $N_1, N_2, \dots, N_k$, then

$$G \cong N_1 \times N_2 \times \cdots \times N_k.$$

Proof. Define

$$\alpha : N_1 \times N_2 \times \cdots \times N_k \longrightarrow G, \quad \alpha(n_1, n_2, \dots, n_k) = n_1 n_2 \cdots n_k.$$

α is bijective. By Lemma 1 (Lecture 10, Lee Lemma 5.2), every element of G can be written uniquely as $n_1 n_2 \cdots n_k$ with $n_i \in N_i$. So α is both injective and surjective.

α is a homomorphism. We must show

$$\alpha((n_1, \dots, n_k)(h_1, \dots, h_k)) = \alpha(n_1, \dots, n_k) \alpha(h_1, \dots, h_k),$$

i.e. $n_1 h_1 n_2 h_2 \cdots n_k h_k = n_1 n_2 \cdots n_k h_1 h_2 \cdots h_k$.

We show we can “bubble” each h_i to the right past the subsequent n_j ’s.

Step 1. Since $N_1, N_2 \trianglelefteq G$ and $N_1 \cap N_2 = \{1\}$ (from condition (3) of the internal direct product), the commuting lemma (Lecture 10, Lee Lemma 5.1) gives $h_1 n_2 = n_2 h_1$. So:

$$n_1 h_1 n_2 h_2 \cdots n_k h_k = n_1 n_2 h_1 h_2 n_3 h_3 \cdots n_k h_k.$$

Step 2. By Theorem 4.5 of Lee, $N_1 N_2 \trianglelefteq G$, and since $(N_1 N_2) \cap N_3 = \{1\}$, the commuting lemma gives $h_1 h_2 n_3 = n_3 h_1 h_2$. Continuing,

$$n_1 n_2 h_1 h_2 n_3 h_3 \cdots n_k h_k = n_1 n_2 n_3 h_1 h_2 h_3 n_4 h_4 \cdots n_k h_k.$$

Repeating this process for all k factors:

$$n_1 h_1 n_2 h_2 \cdots n_k h_k = n_1 n_2 \cdots n_k h_1 h_2 \cdots h_k = \alpha(n_1, \dots, n_k) \alpha(h_1, \dots, h_k).$$

Hence α is a homomorphism, and therefore an isomorphism. □

Remark 1.1: Convention (from lecture)

We henceforth write $G = N_1 \times N_2 \times \cdots \times N_k$ for both internal and external direct products, as the two are isomorphic and context makes the meaning clear.

Example 1.1: $\mathbb{Z}_{15} \cong \mathbb{Z}_3 \times \mathbb{Z}_5$ (from lecture; cf. Lee, Example 5.1)

We show $\mathbb{Z}_{15}$ is the internal direct product of $N_1 = \langle 3 \rangle$ and $N_2 = \langle 5 \rangle$.

Normality. $\mathbb{Z}_{15}$ is abelian, so every subgroup is normal: $N_1, N_2 \trianglelefteq \mathbb{Z}_{15}$.

Explicit subgroups.

$$N_1 = \langle 3 \rangle = \{0, 3, 6, 9, 12\}, \quad N_2 = \langle 5 \rangle = \{0, 5, 10\}.$$

So $|N_1| = 5$ and $|N_2| = 3$.

Trivial intersection.

$$N_1 \cap N_2 = \{0\}.$$

(Since $|N_1| = 5$ and $|N_2| = 3$ are coprime, any common element has order dividing both 5 and 3, hence order 1.)

$N_1 + N_2 = \mathbb{Z}_{15}$. By the product formula (Lee, Thm. 4.4):

$$|N_1 + N_2| = \frac{|N_1||N_2|}{|N_1 \cap N_2|} = \frac{5 \cdot 3}{1} = 15 = |\mathbb{Z}_{15}|.$$

Hence $\mathbb{Z}_{15} = N_1 \times N_2$, so by Theorem 1.1:

$$\mathbb{Z}_{15} \cong \mathbb{Z}_5 \times \mathbb{Z}_3 \cong \mathbb{Z}_3 \times \mathbb{Z}_5. \quad \square$$

2 p -Groups

Definition 2.1: p -element; p -group (from lecture; cf. Lee, Def. 5.3)

Let p be a prime and G a group.

- An element $a \in G$ is called a p -element if $|a| = p^n$ for some integer $n \geq 0$.
- G is called a p -group if every element of G is a p -element.

Note: $n = 0$ gives $|a| = 1$, so the identity is a p -element for every prime p .

Lemma 2.1: Characterisation of finite p -groups (from lecture; cf. Lee, Thm. 5.5 and Cor. 5.2)

A finite group G is a p -group if and only if $|G| = p^k$ for some $k \geq 0$.

Remark 2.1: This uses Cauchy's theorem (from lecture)

The ($\Leftarrow$) direction is straightforward: if $|G| = p^k$, then by Lagrange's theorem every element order divides p^k , hence is a power of p , so every element is a p -element.

The ($\Rightarrow$) direction: if $|G|$ is divisible by some prime $q \neq p$, then by Cauchy's Theorem (Lee, Thm. 5.5 for the abelian case; the general case requires more work), G has an element of order q , which is not a p -element. So $|G|$ must be a power of p .

The lecturer notes this won't be used directly in the following proofs.

Remark 2.2: p -groups need not be cyclic (from lecture)

$\mathbb{Z}_2 \times \mathbb{Z}_2$ is a 2-group (every nonidentity element has order 2), but is not cyclic (it has no element of order $4 = |\mathbb{Z}_2 \times \mathbb{Z}_2|$).

2.1 p -Elements Form a Subgroup

Lemma 2.2: p -elements form a subgroup of an abelian group (Lemma 3 from lecture; cf. Lee, Lemma 5.3)

Let p be a prime and G an abelian group. Let

$$G_p := \{a \in G : a \text{ is a } p\text{-element}\}.$$

Then G_p is a subgroup of G .

Proof. Identity. $|1_G| = 1 = p^0$, so $1_G \in G_p$.

Closure. Let $g, h \in G_p$, so $|g| = p^a$ and $|h| = p^b$ for some $a, b \geq 0$. Let $k = \max(a, b)$. Since G is abelian:

$$(gh)^{p^k} = g^{p^k} h^{p^k}.$$

Since $|g| = p^a \mid p^k$ and $|h| = p^b \mid p^k$, we have $g^{p^k} = h^{p^k} = 1$. Hence $(gh)^{p^k} = 1$, so $|gh|$ divides p^k . Therefore $|gh|$ is a power of p , and $gh \in G_p$.

Inverses. If $g \in G_p$, then $|g^{-1}| = |g| = p^a$, so $g^{-1} \in G_p$.

Hence $G_p \leq G$. □

Remark 2.3: Abelian is essential (from lecture; cf. Lee, p. 89)

Lemma 3 fails for nonabelian groups. In Sym_3 , both (12) and (23) are 2-elements (order 2), but their product $(12)(23) = (132)$ has order 3, which is not a 2-element.

2.2 Decomposing Elements into p -Elements

Lemma 2.3: Elements decompose into p -elements (Lemma 4 from lecture; cf. Lee, Lemma 5.4)

Let G be any group, and let $1 \neq a \in G$ be an element of finite order. Then

$$a = a^{n_1} a^{n_2} \cdots a^{n_k}$$

for some integers $n_1, n_2, \dots, n_k$, where each a^{n_i} is a p_i -element for some prime p_i dividing $|a|$.

Proof. By induction on ℓ , the number of distinct primes dividing $|a|$.

Base case $\ell = 1$: $|a| = p^m$ for some prime p and $m \geq 1$. Then a is itself a p -element, so set $k = 1$ and $n_1 = 1$.

Inductive step $\ell > 1$: Assume the result holds for any element whose order has fewer than ℓ distinct prime divisors. Write

$$|a| = p^m q, \quad \text{where } \gcd(p, q) = 1 \text{ and } p \text{ is prime.}$$

Since $\gcd(p^m, q) = 1$, by Bezout's theorem (Lee, Cor. 2.1) there exist $u, v \in \mathbb{Z}$ such that

$$p^m u + qv = 1.$$

Hence

$$a = a^1 = a^{p^m u + qv} = \underbrace{(a^q)^v}_{p\text{-element}} \cdot \underbrace{(a^{p^m})^u}_{\text{order divides } q}.$$

$(a^q)^v$ is a p -element. We have $(a^q)^{p^m} = a^{qp^m} = 1$, so $|a^q|$ divides p^m , hence a^q is a p -element, and so is $(a^q)^v$.

$(a^{p^m})^u$ has order dividing q . Since $|a^{p^m}|$ divides q , and q has $\ell - 1$ or fewer distinct prime divisors, the inductive hypothesis applies to a^{p^m} . So $(a^{p^m})^u$ is a product of elements of orders that are prime powers, each prime dividing q .

Combining, a is a product of p -elements for primes p dividing $|a|$. □

Remark 2.4: Why this matters (from lecture)

Lemma 4 is the key step that allows us to break any finite abelian group into a direct product of p -subgroups in the next lecture (Lee, Lemma 5.5). Each element of G is a product of p_i -elements, one from each p_i -subgroup G_{p_i} — this is what drives the decomposition $G = G_{p_1} \times G_{p_2} \times \cdots \times G_{p_k}$.

Checklist of Skills

- ☐ Prove Theorem 1.1 (internal $\cong$ external): construct α by concatenation, bijectivity from Lemma 1 (unique expression), homomorphism property via the commuting lemma applied iteratively.
- ☐ Verify $\mathbb{Z}_{15} \cong \mathbb{Z}_3 \times \mathbb{Z}_5$ using the internal direct product: check normality, trivial intersection, product formula for order.
- ☐ State the definition of a p -element and a p -group.
- ☐ State the characterisation: G finite p -group $\iff |G| = p^k$.
- ☐ Give a counterexample to “ p -groups are cyclic”: $\mathbb{Z}_2 \times \mathbb{Z}_2$.
- ☐ State and prove Lemma 3: G_p is a subgroup of G when G is abelian (check identity, closure using abelian property, inverses).
- ☐ Explain why Lemma 3 fails for nonabelian groups (product of two order-2 elements in Sym_3 has order 3).
- ☐ State and prove Lemma 4 by induction on the number of distinct prime divisors of $|a|$, using Bezout’s theorem to split $a = (a^q)^v \cdot (a^{p^m})^u$.
- ☐ Explain how Lemma 4 feeds into the decomposition $G = G_{p_1} \times \cdots \times G_{p_k}$ (coming in Lecture 12).